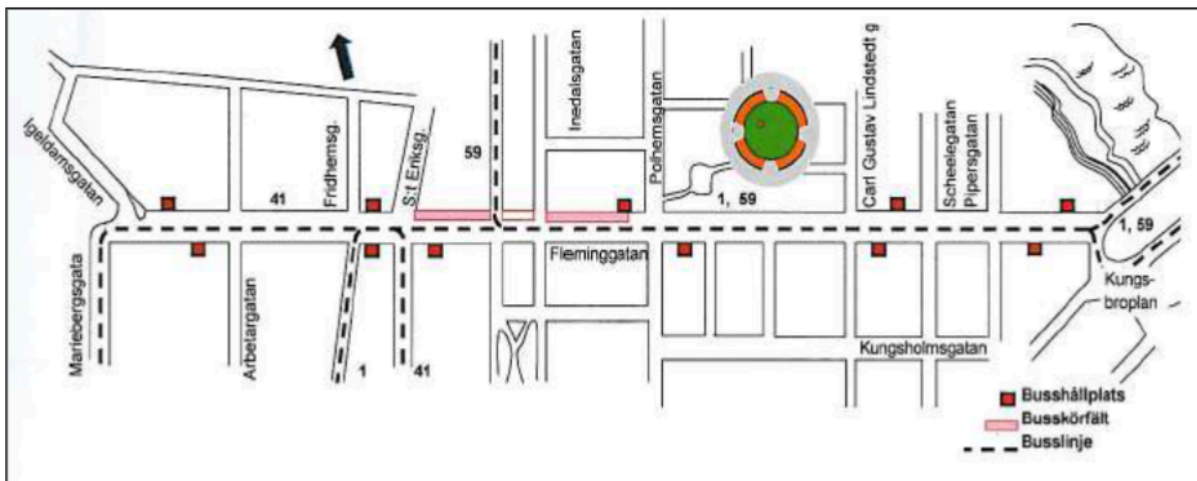


Traffic Signal Design



Course: AH2171 Traffic Engineering and
Management

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1 Introduction

1.1 Background

Traffic signals play a crucial role in managing vehicle and pedestrian flows at intersections, ensuring the safe and efficient sharing of right-of-way among competing road users. However, the implementation of traffic signals introduces delays and stops, which can have a negative impact on overall traffic performance if not properly optimized. Thus the design of signal timings needs to strive for the balance between capacity maximization and delay minimization, taking into account the dynamic nature of traffic demand.

1.2 Objective

This project focuses on the intersection of Scheelegatan and Kungsholmsgatan in Stockholm, Sweden, as shown in Figure 1.1. The aim of the project is to analyse its existing signal operation and propose improvements through data collection, manual calculations and simulation-based evaluation.

The ultimate goal of the project is to enhance the understanding of how signal timing parameters influence traffic behaviour, and to develop a signal plan that improves safety, reduces delay and accommodates varying demand levels.

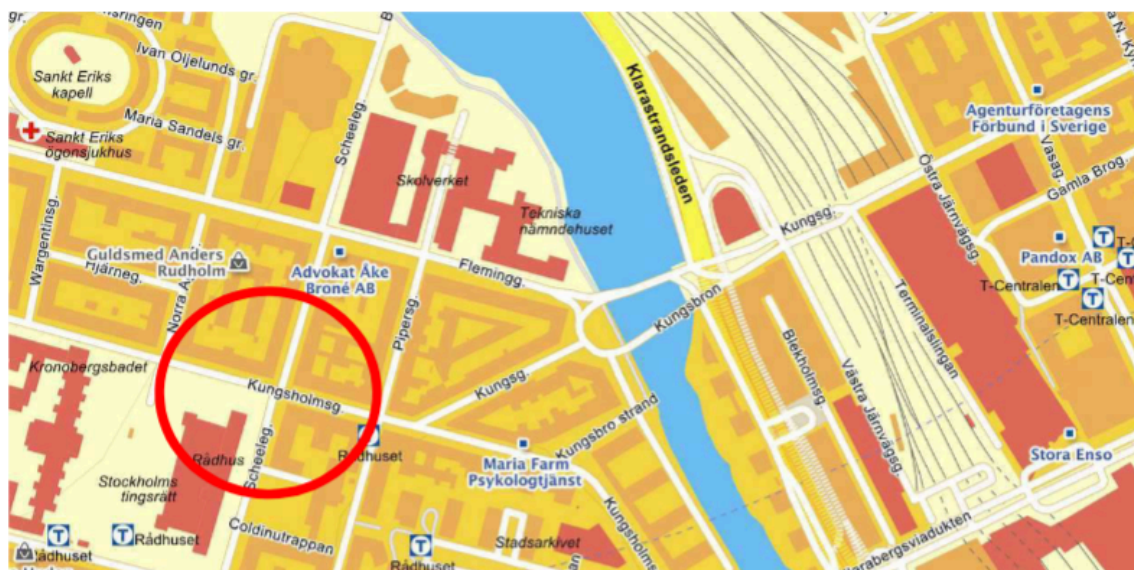


Figure 1.1 Location of the intersection Scheelegatan- Kungsholmsgatan

2 Methodology

2.1 Data Collection

2.1.1 Geometric Design and Signal Time Setting

The measurements were carried out in groups for each approach according to the grouping and time schedule, which included the following tasks:

- 1) Measurement of the approaches geometry (lane configuration, lane widths, length of the short lane, length / width of mid refuges, length of pedestrian crossings etc. information required to make a sketch of the intersection).
- 2) The intersection's signal phase images (i.e. which approaches and pedestrians crossing places that have green in respective Phase).
- 3) Measurement of the signaling cycle time and green time for each signal phase.
- 4) Measure and record cycle times and green times for at least five cycle times.

2.1.2 Traffic Flow

Manual counting of the following modes for one hour (13:00-14:00) according to the grouping arrangement.

- 1) Registration of the number of vehicles passing the stop of the studied approach considering the driving direction (L, T, R) and lanes.
- 2) Registration of the number of bicycles crossing the stop line in each approach (not per lane).
- 3) Registration of the number of pedestrian traffic crossing the centre island (both directions) for each crossing.

2.1.3 Saturation Flow

Register the flow of each approach under saturated queue dissipation's state after the start of the green light for each cycle time as follows:

- 1) Time of passage of the stop line of the fourth vehicle in the lane (i.e. Queue position = 4). Time is normally 00:00 if the stopwatch is started at this event in each cycle.

- 2) Time of passage of the last vehicle standing still in the queue at the green time start-Queue position for the latter vehicles.

Saturation flow measurements for each lane were performed by two people as follows:

- 1) Observer 1 identifies which vehicle is the last in line when the green period starts and shouts out that vehicle's queue position when the vehicle's front wheel passes the stop line.
- 2) Observer 2 will start the stopwatch when the front wheels of the vehicle with queue position 4 crosses the stop line and stop the stopwatch when observer 1 calls out according to above.
- 3) The observer records the result of the current signal cycle in tabular form.

2.1.4 Cruise Time

Measurements of travel time for vehicles passing the south approach of Scheelegatan driving through the intersection towards Fleminggatan. The travel time for vehicles that meet

a green signal at the Scheelegatan intersection and a green signal at the downstream intersection, Fleminggatan intersection, is measured.

2.2 Data Processing

The signal timings for the Scheelegatan-Kungsholmsgatan intersection were manually calculated to analyze their impact on intersection performance and to gain insights into the effects of different timing configurations. The geometry layout of the intersection is illustrated in Figure 2.1, while the corresponding traffic data for the off-peak period are provided in Figure 2.2.

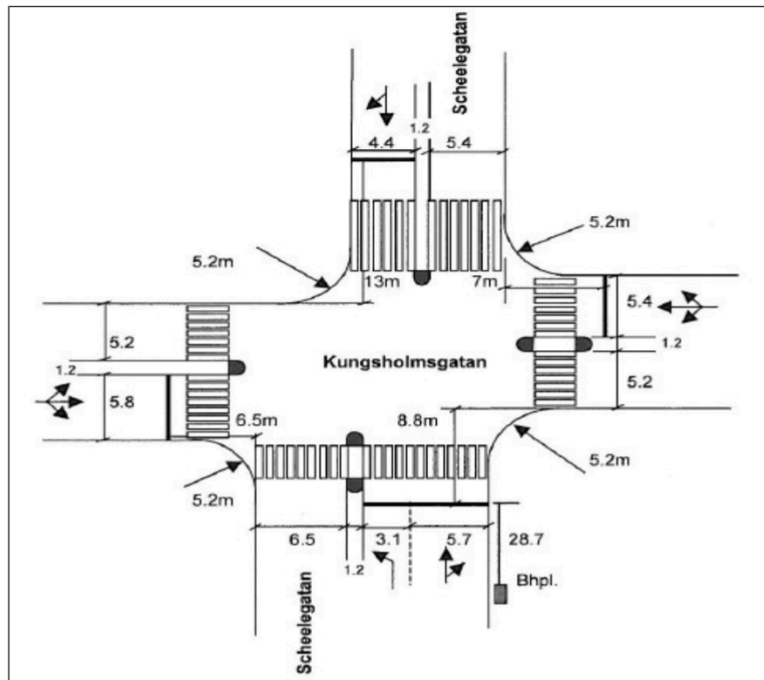
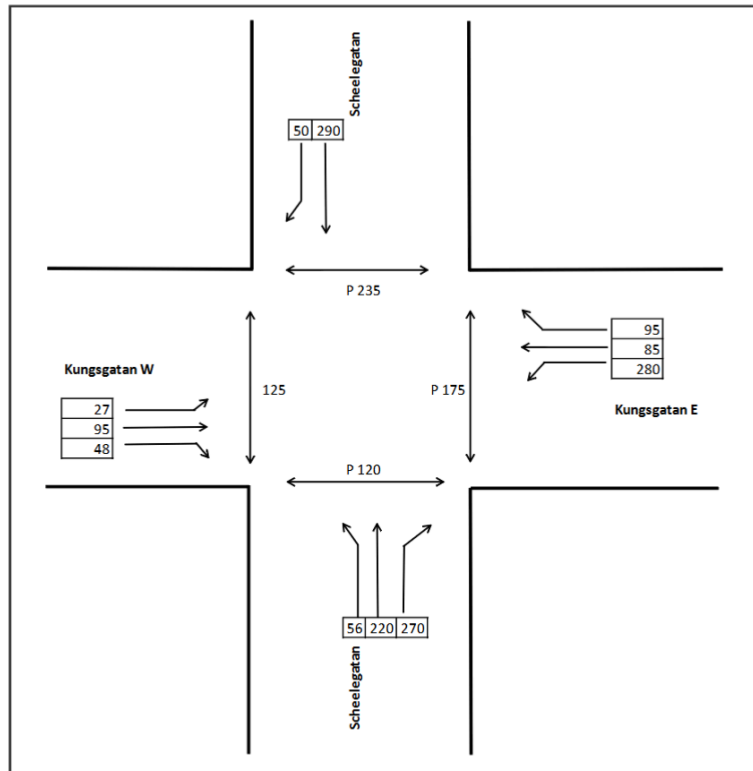


Figure 2.1 Intersection Geometry in Scheelegatan – Kungsholmsgatan



**Figure 2.2 Turning Traffic Movements in Scheelegatan – Kungsholmsgatan
(P=Pedestrians)**

Rough Estimation of Signal Timings

Assuming that the intersection uses 2-stage control, the following rule of thumb can be used to make a rough estimation of the signal timings at the intersection:

- 1) Two green seconds per vehicle and lane.
- 2) Green split is proportional to traffic flows.
- 3) Lost time of approximately 5 seconds per stage change in a small intersection.
- 4) Cycle time not exceeding 100 seconds in a city center.

Based on the rule of thumb above, the signal times for the intersection are estimated by the following steps:

- 1) Determine the signal groups for each stage.
- 2) List the traffic flows for each lane of the intersection.
- 3) Calculate the green times for each lane.
- 4) Use the following formula to calculate the signal timing:

$$t_{Est.} = t_{g1} + t_{g2} + \frac{3600}{100} \cdot L$$

Where:

t_{g1} is the maximum green time of the 1st stage in one hour;

t_{g2} is the maximum green time of the 2nd stage in one hour;

L is the lost time per cycle.

- 5) If the calculated signal timing is less than one hour (3600 secs), then there is enough capacity with the number of lanes at the intersection.

Saturation Flow Calculation

To determine the saturation flow for each lane group, first identify the movement type involved—such as through movements, right turns, left turns, or any combined movement. Once the movement type is defined, select the appropriate empirical Swedish saturation flow value recommended for that movement. If the movement involves secondary conflicts, adjustments must be made accordingly. In cases where left turns conflict with opposing traffic, the saturation flow is reduced by the opposing traffic volume, calculated as:

$$S_{LT} = S_{total} - q_{opposing}$$

Optimal Cycle Time Calculation

To calculate the optimal cycle time using Webster's formula, begin by determining the total lost time L , which is found by multiplying the standard 5-second lost time per stage change by the total number of stage changes in the signal cycle. Next, compute the critical flow ratio for each stage by dividing the traffic flow by the corresponding saturation flow, $(q/s)_i$, and then sum these ratios across all stages. With the total lost time and the sum of critical ratios, apply Webster's formula,

$$C_{opt} = \frac{1.5 \cdot L + 5}{1 - \sum \frac{q_i}{s_i}}$$

to obtain the optimal cycle length. Finally, round the resulting value to a practical whole number suitable for implementation in an urban signal timing plan.

Optimal Green Splits Calculation

To determine the optimal green splits, start by using the previously calculated cycle time C and total lost time L . For each stage, compute the flow ratio by dividing the traffic flow by its corresponding saturation flow, q_i/s_i , and then sum all stage flow ratios. The available effective green time, $C-L$, is then distributed proportionally among the stages based on their relative flow ratios. This is done using the formula

$$g_i = \frac{\frac{q_i}{s_i}}{1 - \sum \frac{q_i}{s_i}},$$

which assigns green time in proportion to each stage's contribution to overall demand. Finally, verify that each resulting green interval satisfies the minimum green time requirements to ensure safe and efficient signal operation.

Minimum Green Calculation

For vehicle movements, apply the Swedish standard minimum green time of 4 seconds when a stop-line detector is present. For pedestrian movements, begin by measuring the length of each pedestrian crossing, L , and then calculate the minimum required pedestrian green time using the formula $G_g = L/Vfg$, based on a walking speed of 1.2 m/s. The resulting value should be rounded up to the nearest whole second to ensure sufficient time for pedestrians to safely cross the roadway.

Calculation of Inter-green Times

To maintain safety during signal phase transitions, inter-green times (red intervals) were calculated for the critical conflict points in the intersection. The safety time R represents the period during which all approaches must remain red to allow a clearing vehicle or pedestrian to safely leave the conflict zone before a conflicting movement receives a green signal.

The calculations were performed according to the standard formula for signal timing:

$$R = (S_u / V_u) - (S_f / V_f) - RG$$

Where the following standard parameters were applied:

S_u (Clearance length): The distance from the stop line to the end of the conflict zone. For vehicles, a standard vehicle length of 6 meters was added to this distance.

V_u (Clearing speed): 10 m/s for vehicles and 1.4 m/s for pedestrians.

S_f (Approaching length): The distance from the entering stop line to the start of the conflict zone. For pedestrians, this value is set to 0.

V_f (Approaching speed): 14 m/s (50 km/h) for entering vehicles.

RG (Red-amber period): Fixed at 1.5 seconds.

2.3 Data Analysis

2.3.1 Network modeling in TRANSYT

The TRANSYT tool was used to model the intersection at Scheelegatan-Kungsholmsgatan. The network model incorporated key data elements, including: intersection geometry, traffic flows and turning movements, saturation flows, signal stages and intergreen times.

A step-by-step approach was followed to construct the network:

1. Turning movements and traffic flows were input for all approaches.
2. The saturation values were determined on the basis of geometric traffic conditions. These can be different at each location.
3. The intergreen matrix and signal stages were configured to reflect realistic traffic conditions.

Evaluation of Signal Plans

Two evaluations were performed:

1. Hand Calculations: The calculated optimal cycle time and green splits were input into TRANSYT. Performance indicators such as total network delay, network performance index (PI), and average speed were recorded.
2. Optimized Signal Plan: TRANSYT's built-in optimization tool was used to adjust offsets and green splits. The results were compared to the hand-calculated signal plan to assess potential improvements in performance metrics.

Sensitivity Analysis on Traffic Demand

The network was subjected to varying traffic conditions to evaluate its capacity under increased flow scenarios. This includes increasing traffic flow to approach capacity (10–20%), and exceeding capacity (20–30%).

Each scenario was tested with different cycle times (e.g., 80, 100, and 120 seconds), and results such as total delay, PI, and average speed were compared to identify trends and system limitations.

Graphical and Qualitative Analysis

Cycle time versus performance indicators, such as uniform delay and random delay, were plotted to visualize system behavior. Additionally, qualitative assessments were conducted to propose design changes when the system approached or exceeded capacity.

2.3.2 Analysis of Signal Timings with Microscopic Simulation in VISSIM

VISSIM Network Configuration

The microscopic traffic simulation tool VISSIM was employed to evaluate the intersection's signal timings. Two predefined network files were used:

1. Kungsholmsgatan_1.inpx: Reflecting the calculated signal timings.
2. Kungsholmsgatan_2.inpx: Representing real-world traffic conditions with updated traffic flows.

Signal timings, including optimal green splits, cycle times, and intergreen periods from section 4.3.2, were input into the first network file. The simulation was then run to observe intersection performance under idealized conditions.

Performance Evaluation

Key performance metrics, such as delay per approach and overall uniform delay, were extracted for both vehicle and pedestrian traffic. These were compared with TRANSYT results to validate the consistency of the findings across tools.

Adjustment and Optimization

Using VISSIM's animation features, potential inefficiencies in the signal timings were identified. Adjustments to green times and phase sequences were proposed and tested iteratively. Improved signal plans were then implemented and reevaluated.

Vehicle-Actuated Control

The provided vehicle-actuated control logic was tested by modifying maximum extension times for each stage. Different settings were analyzed to observe the impact on signal behavior and performance compared to fixed-time control. Outputs, such as delays and average speeds, were recorded to assess the viability of dynamic control.

3 Results

Stage A consists of the East–West directions having the green light, while Stage B consists of the North–South directions having the green light.

3.1 Manual calculations

3.1.1 Rough Calculations

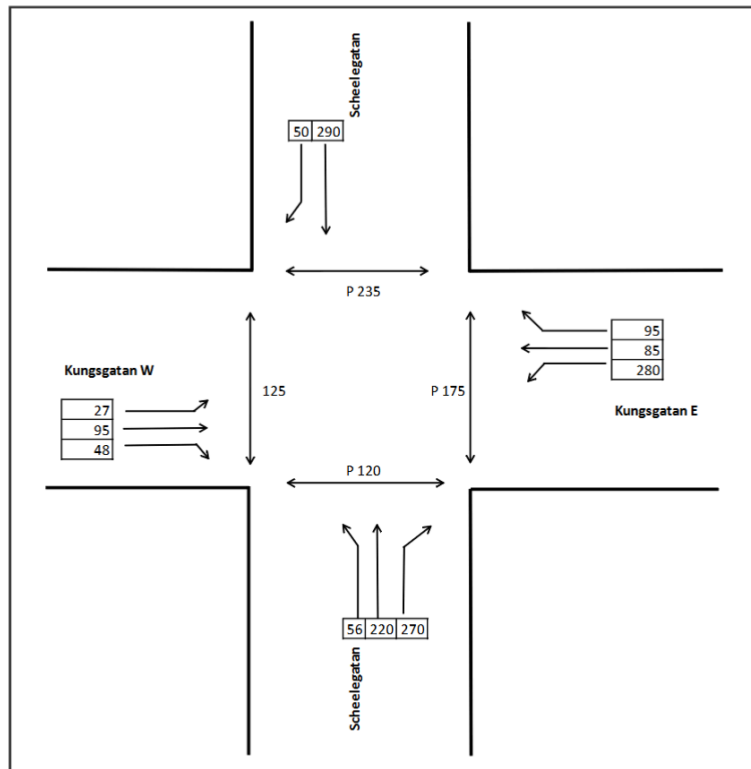


Figure 3.1 Turning Traffic Movements in Scheelegatan – Kungsholmsgatan (P=Pedestrians)

Figure 3.1 shows that the Kungsgatan East lane and the straight-and-right-turn lane of Scheelegatan South carry the highest traffic volumes, with 460 and 490 vehicles per hour, respectively.

Q1. Is there enough capacity?

$$t_{Est.} = 980 + 920 + 36 * 10 = 2260s < 3600s$$

Yes, there is enough capacity with the current number of lanes.

Q2. Is there enough capacity to increase the traffic to twice the observed flows?

$$t_{Est.} = 980 * 2 + 920 * 2 + 36 * 10 = 4160s > 3600s$$

There is thus not enough capacity with the number of lanes in the current layout.

Q3. If not, where are extra lanes needed?

Since Kungsgatan East now carries a big volume, an extra left-turning lane was added at Kungsgatan East. Now the new left turning lane and the straight-and-right-turn lane of Scheelegatan South carry the highest traffic volumes, with 560 and 980 vehicles per hour, respectively.

$$t_{Est.} = 980 * 2 + 560 * 2 + 36 * 10 = 3440s < 3600s$$

There would be enough capacity to increase the traffic to twice the observed flows if an extra left-turning lane at Kungsgatan East would be added.

3.1.2 Detailed Calculations

The calculated saturation flows and the critical ratios of flows to their saturation flows are as shown in Table 3.1.

Table 3.1 Calculation for saturation flows and their critical ratios

Direction	Demands(veh/h)	Saturation flows(veh/h)	Critical Ratios
Scheelgatan North	340	1600	0.21
Kungsgatan West	170	1270	0.13
Scheelgatan South-Left turn	56	1110	0.05
Scheelgatan South-Right turn&Straight	490	1600	0.31
Kungsgatan East	460	1307	0.35

From Table 3.1, it is evident that the ratios for Scheelgatan South—Right turn & Straight and Kungsgatan East are the highest within their respective stages. Consequently, the critical ratios are 0.31 for Stage B and 0.35 for Stage A.

With this, optimal cycle time is calculated to be 58.8 seconds, which is rounded up to 59 seconds.

The optimal green split of stage A is calculated to be 25.9 seconds while that of stage B is 22.5 seconds.

The minimum green period for all pedestrian crossings in the intersection Scheelegatan – Kungsholmsgatan is as shown in Table 3.2.

Table 3.2 Calculation for the minimum green period for pedestrian crossings

Pedestrian crossing	Minimum green period (s)
North crossing	9.2
West crossing	10.2
South crossing	13.8
East crossing	9.8

The calculated red periods with respect to approaching and clearing pedestrians, and vehicle movements in the intersection Scheelegatan– Kungsholmsgatan are as shown in Table 3.3. The inter-green conflict matrix is as shown in Table 3.4.

Table 3.3 Calculation of Critical Inter-green Times

Change From → To	Conflict Description	Type	Su (m)	Sf (m)	Final R (s)
Stage B → Stage A	<i>(Scheelegatan stops → Kungsholmsgatan starts)</i>				
	Vehicle SB vs. Vehicle EB	Vehicle	31.2	6.5	1.2
	Pedestrian West vs. Vehicle EB	Pedestrian	5.8	0	2.6
Stage A → Stage B	<i>(Kungsholmsgatan stops → Scheelegatan starts)</i>				
	Vehicle EB vs. Vehicle NB	Vehicle	29	8.8	0.8
	Pedestrian South vs. Vehicle SB	Pedestrian	8.8	0	4.8

Table 3.4 Inter-green Conflict Matrix (seconds)

Terminating \ Starting	Veh N	Veh S	Ped E	Ped W	Veh E	Veh W	Ped N	Ped S
Veh N (Stage B)	-	-	X	X	1.2	1.2	-	-
Veh S (Stage B)	-	-	X	X	1.2	1.2	-	-
Ped E (Stage B)	X	X	-	-	2.6	2.6	-	-
Ped W (Stage B)	X	X	-	-	2.6	2.6	-	-
Veh E (Stage A)	0.8	0.8	-	-	-	-	X	X
Veh W (Stage A)	0.8	0.8	-	-	-	-	X	X
Ped N (Stage A)	4.8	4.8	-	-	X	X	-	-
Ped S (Stage A)	4.8	4.8	-	-	X	X	-	-

The proposed signal timings are as shown in Table 3.5. The detailed signal diagram can be found in Figure 3.2.

Table 3.5 Signal timing for the intersection

	Green period (s)	Red period (s)	Inter-green time (s)
Stage A	25.0	20.4	5
Stage B	20.4	25	4.5

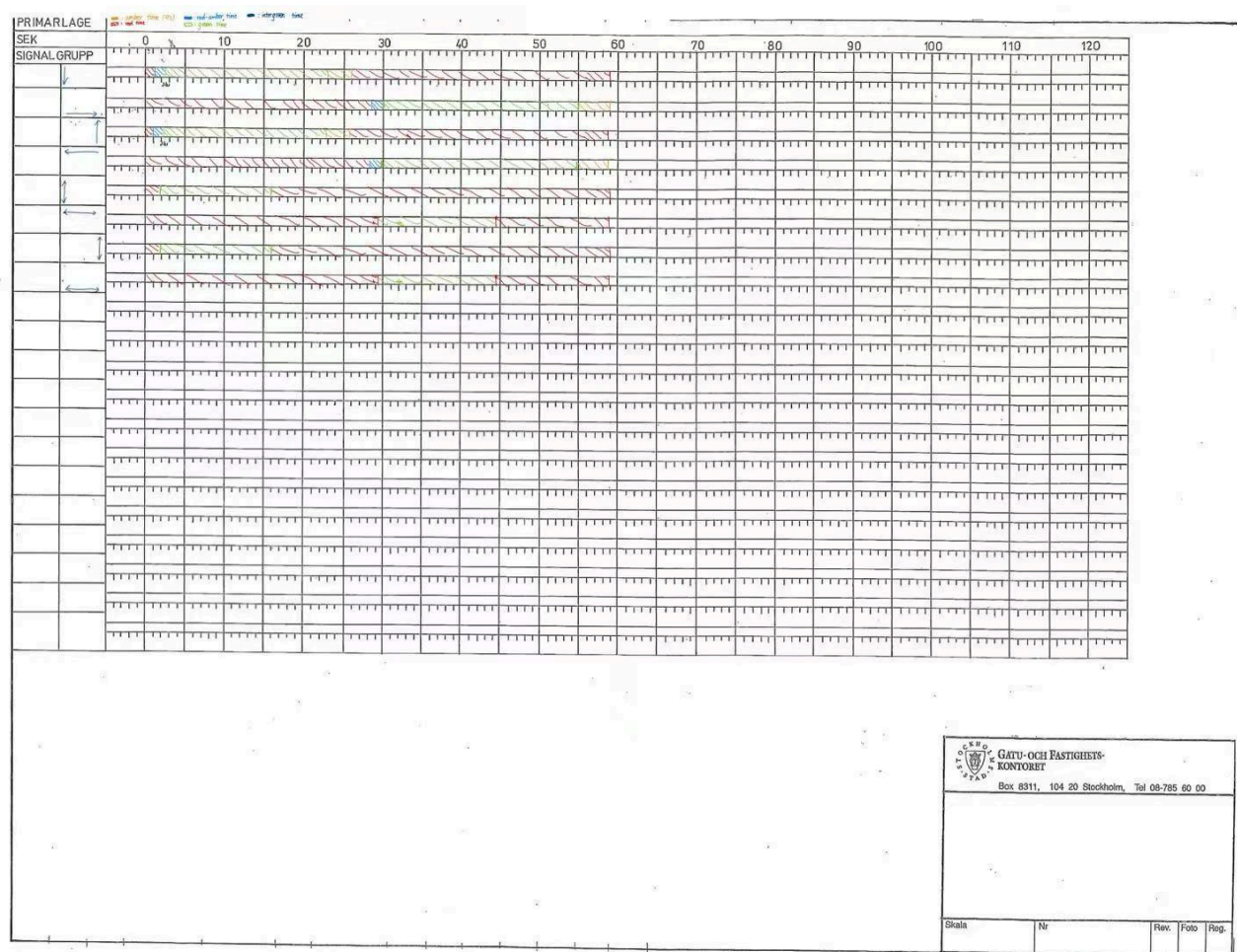


Figure 3.2 Signal diagram

3.2 Transyt

This chapter presents the results from the TRANSYT simulations, including the base case evaluation, optimization, sensitivity analysis, and network modifications.

3.2.1 TRANSYT Model Evaluation (Isolated Case)

The intersection was first modelled using the manually calculated timings (Cycle time: 59s) and then optimized using TRANSYT's internal optimizer (CYOP and Full Run).

Table 3.6: Comparison between Manual and Optimized Signal Plans

Performance Metric	Hand Calculation	Optimized Plan
Cycle Time (s)	59	50
Total Network Delay (PCU-hr/hr)	9.34	8.1
Performance Index (£ per hr)	149.32	132.16
Network Average Speed (km/h)	12.9	13.95
Highest Degree of Saturation (%)	87.99	79.99

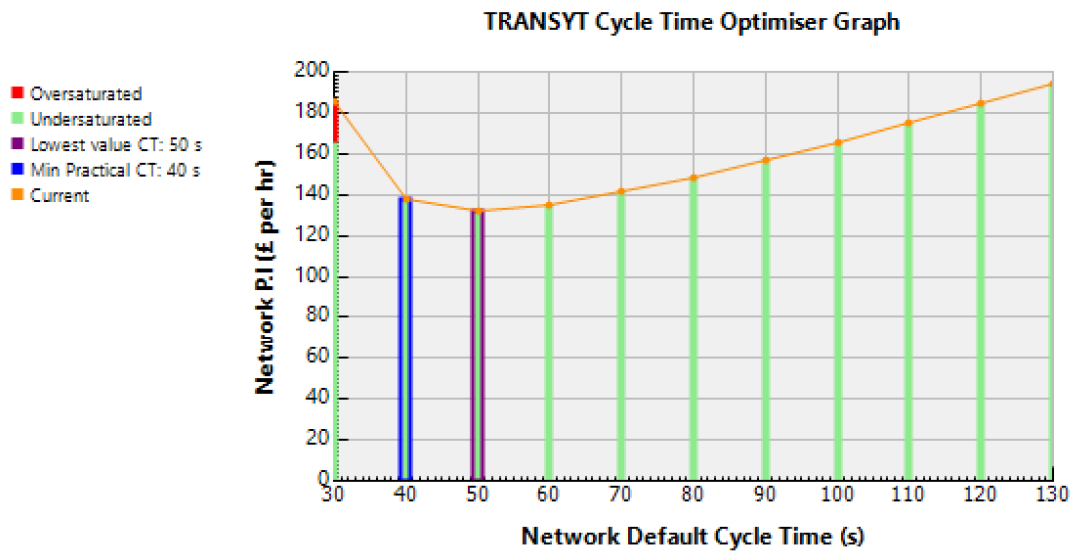


Figure 3.3: TRANSYT Cycle Time Optimization (Base Scenario).

This graph illustrates the relationship between the network cycle time and the Performance Index (PI). The optimization results indicate an optimal cycle time of 50 seconds for the existing intersection layout. As shown in the curve, cycle times shorter than 50 seconds cause a sharp increase in PI due to capacity constraints (oversaturation), while extending the cycle time beyond the optimum leads to a gradual increase in PI caused by unnecessary delays.

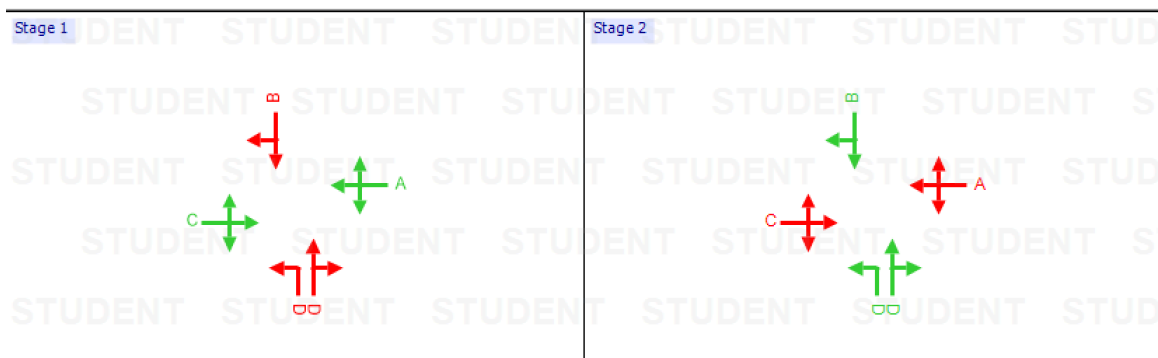


Figure 3.4: Original 2-Stage Phasing Diagram (Base Scenario).

This diagram illustrates the initial signal control strategy consisting of two stages: East-West (Stage A) and North-South (Stage B). In this configuration, left-turning traffic from Kungsholmsgatan runs concurrently with the through traffic and must yield to opposing vehicles. This layout was optimized in TRANSYT with a cycle time of 50 seconds.

Cycle Time Analysis The following graphs visualize how different performance metrics change with varying cycle times (from 30 to 150 seconds).

The graph (Figure 3.1) illustrates the relationship between the network cycle time and the uniform delay. The X-axis represents the cycle time in seconds, ranging from 30 to 150 seconds. The Y-axis displays the uniform delay measured in PCU-hr/hr.

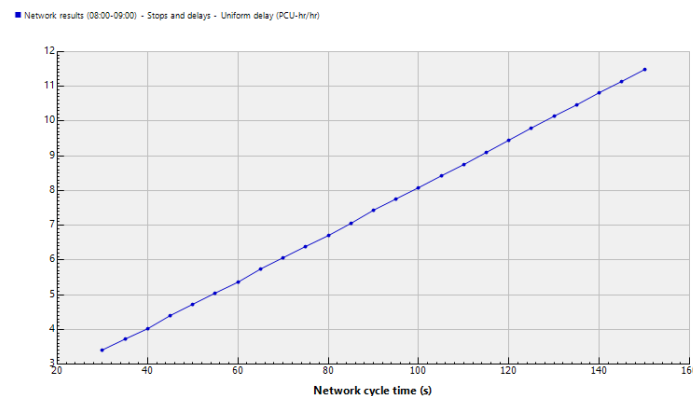


Figure 3.5: Uniform Delay vs. Cycle Time

This graph (figure 3.2) shows the random and oversaturated delay relative to the cycle time. The X-axis shows the cycle time in seconds, and the Y-axis represents the delay in PCU-hr/hr.

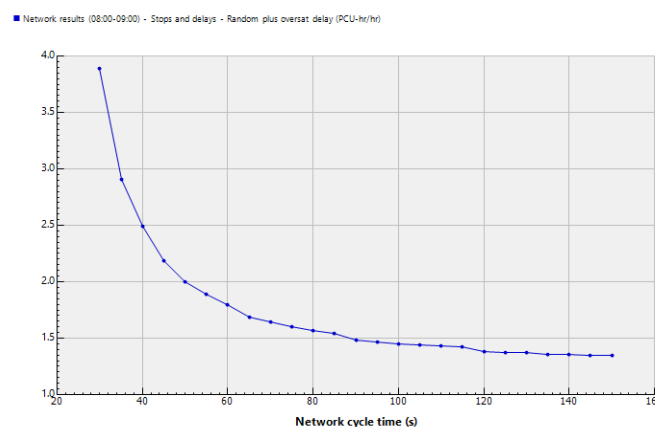


Figure 3.6: Random + Oversaturated Delay vs. Cycle Time

The graph (Figure 3.3) displays the Performance Index (PI) as a function of the network cycle time. The X-axis represents the cycle time (seconds), and the Y-axis represents the PI value (cost per hour).

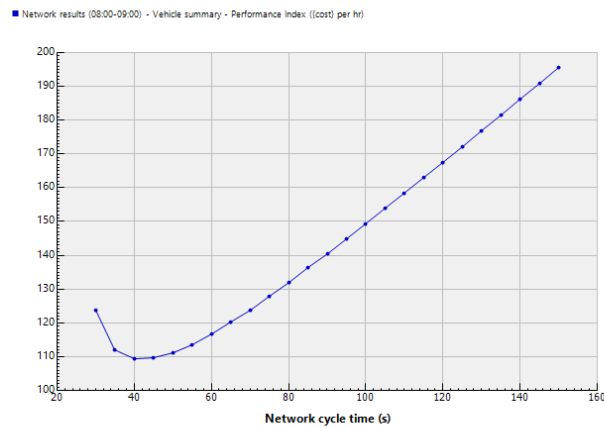


Figure 3.7: Performance Index (PI) vs. Cycle Time

3.2.2 Sensitivity analysis on traffic demand

To test the robustness of the signal plan, the traffic demand was increased by 15% and 30% while maintaining the optimized settings.

Table 3.7: Sensitivity Analysis Results

Performance Metric	100% Flow (Base Case)	115% Flow (Increased)	130% Flow (High)
Cycle Time (s)	50	70	100
Total Network Delay (PCU-hr/hr)	8.1	13.12	23.48
Performance Index (£ per hr)	132.16	206.78	358.47
Network Average Speed (km/h)	13.95	11.459	8.42
Highest Degree of Saturation (%)	79.99	85.85	93.533
Within Capacity (<90%)?	YES	YES	NO

3.2.3 Network modification (Stage 3)

Since the network exceeded capacity at 130% flow, the intersection design was modified. A separate left-turning lane and a third signal stage were introduced for the Westbound left-turn movement (traffic coming from Arm 2/East turning South) to eliminate conflicts with opposing traffic.

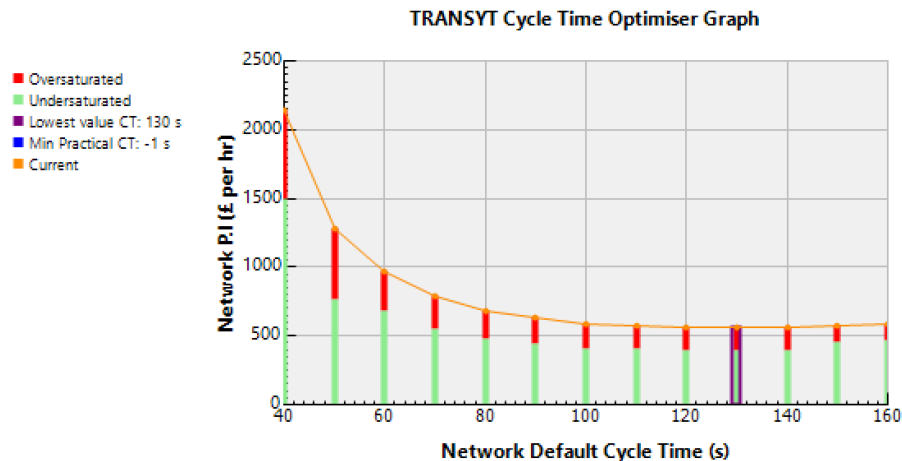


Figure 3.8: TRANSYT Cycle Time Optimization (Modified Layout with 3 Stages).

Following the introduction of a third signal stage to separate the left-turning movement, the optimal cycle time shifts significantly to **130 seconds**. The additional "lost time" introduced by the extra stage change reduces the effective green time available, necessitating a longer total cycle length to maintain capacity and minimize the Performance Index.

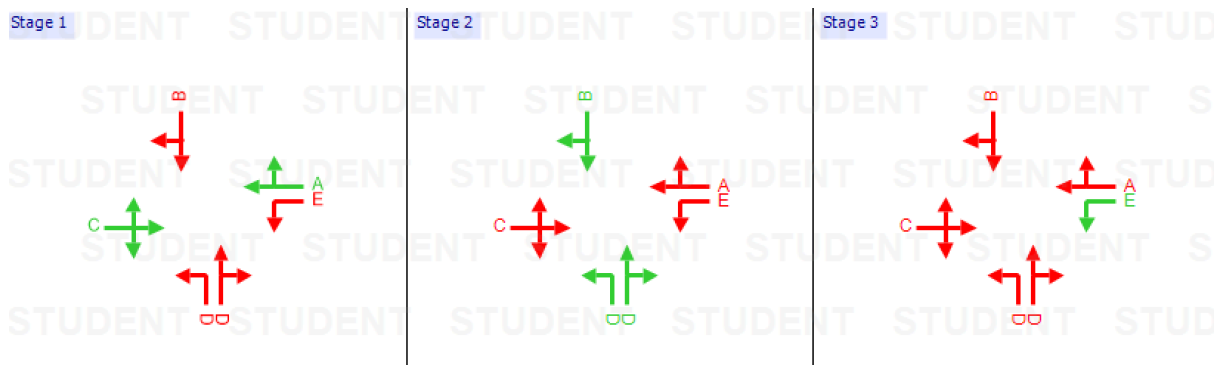


Figure 3.9: Modified 3-Stage Phasing Diagram.

To address safety concerns and conflicts with opposing traffic, a dedicated **Stage 3** was introduced for the Westbound left-turn movement. As shown in the diagram, this separates the turning traffic from the through traffic. However, the introduction of this third stage increased the "lost time" per cycle, requiring the optimal cycle time to be increased to **130 seconds** to handle the demand.

Table 3.8: Results of Network Modification

Performance Metric	Original Layout	Modified Layout	Change / Comment
Configuration	2 Stages	3 Stages (Added WB Left-Turn)	New stage added for safety/capacity.
Cycle Time (s)	100 s	130 s	Drastic increase to handle load.

Highest DOS (%)	93.53%	95.28%	Slightly increased
Total Network Delay (PCU-hr/hr)	23.48	37.7	Increased significantly.
Network Average Speed (km/h)	8.4	5.9	Reduced speed due to longer waiting times.
Performance Index (£ per hr)	358.78	562.21	Higher cost due to increased delays.
Within Capacity?	NO (>90%)	NO (>90%)	Still overloaded despite modification.

3.2.4 Cycle time analysis (Fixed flows)

Finally, different fixed cycle times (80, 100, 120s) were tested against varying traffic flows to analyze if a longer cycle time could handle increased demand without modifying the intersection geometry.

Table 3.9: Impact of Cycle Time on Saturation and Delay

Cycle Time \ Flow	100% Flow	125% Flow	135% Flow
80 seconds	DOS: 76.10% Delay: 9.40 PCU-hr/hr	DOS: 92.62% Delay: 18.92 PCU-hr/hr	DOS: 100.03% Delay: 32.81 PCU-hr/hr
100 seconds	DOS: 75% Delay: 10.65 PCU-hr/hr	DOS: 90% Delay: 19.43 PCU-hr/hr	DOS: 97% Delay: 29.94 PCU-hr/hr
120 seconds	DOS: 75% Delay: 11.97 PCU-hr/hr	DOS: 89% Delay: 20.67 PCU-hr/hr	DOS: 97% Delay: 29.78 PCU-hr/hr

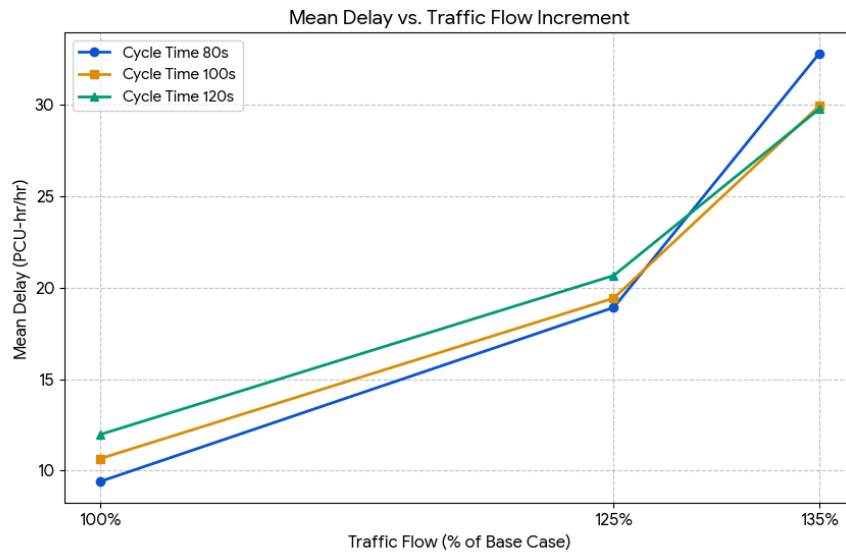


Figure 3.4: Average Delay at Increasing Traffic Flows

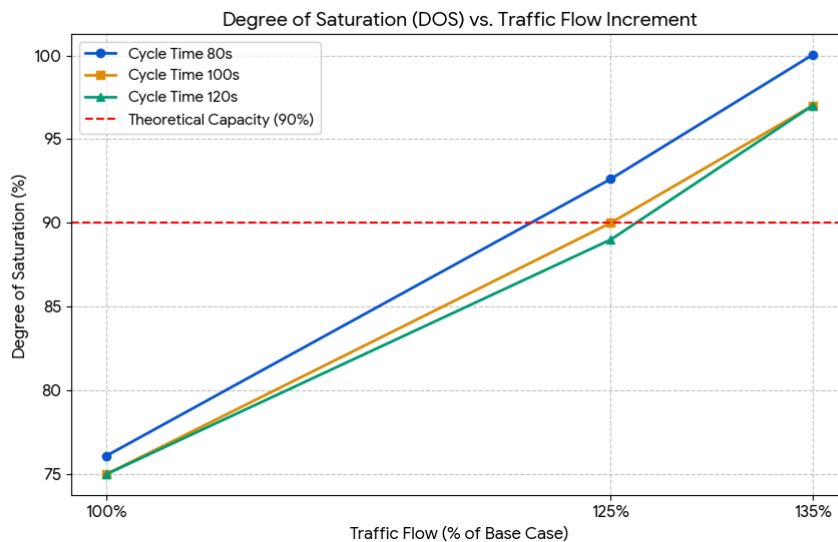


Figure 3.5: Degree of Saturation at Increasing Traffic Flows

3.3 Microscopic simulation - VISSIM

3.3.1 Evaluation of signal plan in VISSIM

For the evaluation using VISSIM, the signal timing plan was adjusted accordingly so that the optimum cycle of 59 seconds was maintained, as shown in Figure 3.4.

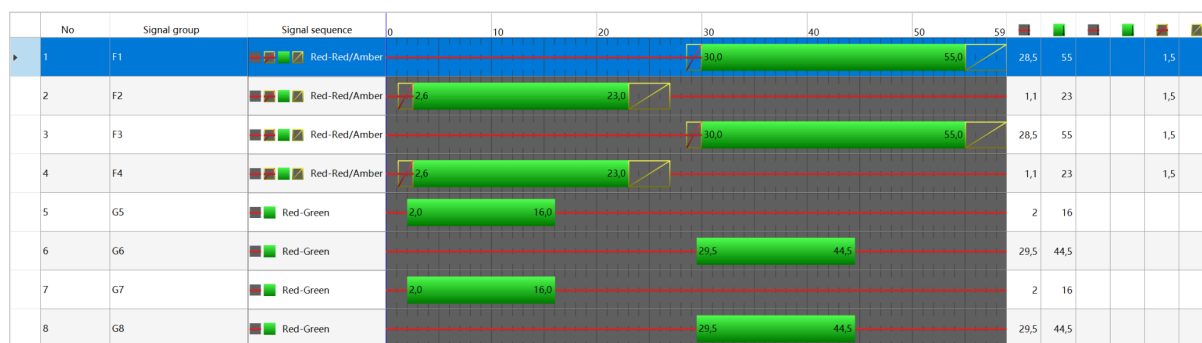


Figure 3.4 VISSIM- Signal Plan

The initial assessment of the calculated signal timings was carried out using the microscopic simulation tool VISSIM. This evaluation measured the delays experienced by both vehicles and pedestrians at each approach, as well as the overall uniform delay for the intersection. The results of the average delays for each direction are as shown in Table 3.10. The pedestrian delays are 0 because there was no pedestrians in the network.

Table 3.10 Average delays in each direction

Direction	Vehicle Delay (s)	Pedestrian Delay (s)
North	18.1	0
West	12.7	0
South	39.4	0
East	32.0	0
Average Delay (s)	29.8	0

3.3.2 Average delays of changed traffic conditions

The simulation was repeated under modified traffic conditions to better represent real-world variations. The updated traffic flows altered intersection performance, producing an average vehicle delay of 35.2 seconds while average pedestrian delay is 0 seconds. Detailed delays are as shown in Table 3.11.

Table 3.11 Average delays with changed conditions

Direction	Vehicle Delay (s)	Pedestrian Delay (s)
North	13.2	0
West	102.0	0
South	34.5	0
East	17.8	0
Average Delay (s)	38.1	0

3.3.3 Improved signal timings

Reduced cycle time

Since the optimal cycle time estimated by TRANSYT was shorter than the manual calculation, we tested reduced cycle times of 50 seconds and 40 seconds. The most significant improvement was observed when shortening the cycle time from 59 seconds to 50 seconds, as shown in Table 3.12.

Table 3.12 Comparison between different cycle time

Cycle time (s)	Vehicle Delay (s)	Pedestrian Delay (s)
59	38.1	0
50	37.2	0
40	37.1	0

Changed green splits

From the animation, it is evident that a long queue develops on Scheelgatan South. To address this, we extended the green time for Stage B while keeping the overall cycle time constant and adjusting the green splits accordingly, as shown in Figure 3.6. As shown in Table 3.13, this modification leads to a reduction in average vehicle delay.

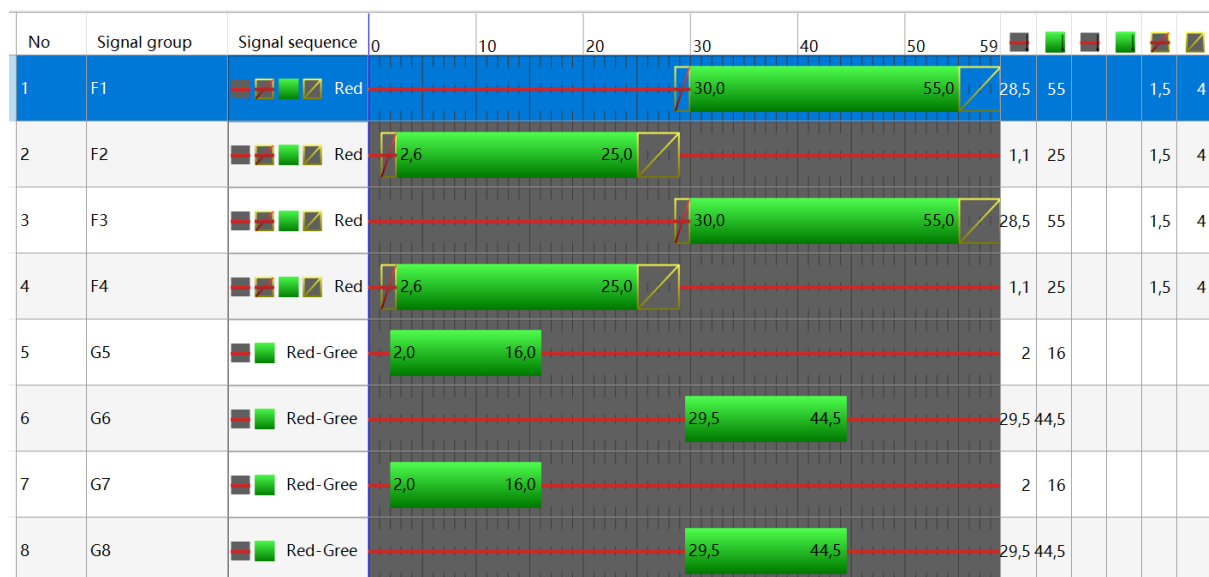


Figure 3.5 VISSIM- Signal Plan with adjusted green splits

Table 3.13 Comparison between different green times

Green time	Vehicle Delay (s)	Pedestrian Delay (s)
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Base	38.1	0
Extended for Stage B	37.2	0

3.3.4 Vehicle actuated control

Table 3.14 presents the vehicle delays under vehicle-actuated control, while the pedestrian delays are 0 because no pedestrians entered. The analysis considers maximum extension times ranging from 20 to 35 seconds, with increments of 5 seconds.

Table 3.14 Vehicle delays with different maximum extension time

Direction	Maximum extension time (s)			
	20	25	30	35
North	12.5	16.3	16.4	16.6
West	11.4	14.5	14.3	15.1
South	46.5	22.0	22.3	21.9
East	42.7	33.5	31.4	33.0
Average Delay (s)	25.9	23.4	22.8	23.3

At a maximum extension time of 30 seconds, the average vehicle delays reach their lowest values. This indicates that the optimal extension time is neither simply the longest nor the shortest, but rather depends on the prevailing traffic flow conditions.

4 Discussion

4.1 Manual calculation compared with signal timing from the street

4.1.1 Comparing the red periods

The primary difference is that the documented longest red phases for pedestrians and vehicle movements exceed the calculated values. This discrepancy is likely due to the implementation of longer red phases in practice to ensure traffic safety.

4.1.2 Comparing with the signal timings in signal plan P2

Several differences can be observed between the signal plan P2 and the calculated results. The cycle time in P2, at 67 seconds, is longer than the calculated cycle time of

approximately 60 seconds. Moreover, P2 incorporates significantly longer red phases for vehicles, which is likely intended to minimize conflicts between pedestrian and vehicle movements. Another notable difference lies in the allocation of green times: while the calculated timings assign equal green durations to signal groups within the same stage, P2 provides a much shorter green time to signal group 1 (west to east) compared to signal group 3 (east to west). This adjustment reflects the considerably lower traffic volume on Kungsholmsgatan West relative to Kungsholmsgatan East, ensuring that green time distribution better matches actual traffic demand.

4.2 Transyt

4.2.1 Evaluation of signal plans: manual vs. optimized

The initial evaluation compared the manually calculated signal timings with the optimized timings produced by TRANSYT. The manual calculation, based on Webster's method, resulted in a 59-second cycle. In contrast, the TRANSYT optimization suggested a shorter 50-second cycle as the most efficient option for the base traffic demand.

The results show that the optimized plan improved the intersection's performance. The Total Network Delay decreased from 9.34 to 8.1 PCU-hr/hr, and the Performance Index (PI) was reduced from 149.32 to 132.16. This indicates that, for the current traffic volumes, a shorter cycle time helps reduce unnecessary waiting, as the intersection still has enough capacity (the highest DOS was only 79.99%).

4.2.2 Analysis of cycle time trends

The graphical analysis of cycle times and performance indicators shows clear patterns:

Uniform Delay:

The results indicate that uniform delay increases steadily as the cycle time becomes longer. This is expected, because vehicles that arrive during the red phase must wait longer when the signal cycle is long.

Random + Oversaturated Delay:

In contrast to uniform delay, this component decreases quickly as the cycle time increases from a very low value. When cycle times are short, a large share of the cycle is lost to intergreens, which reduces the available capacity. If the capacity is close to or below the demand, small variations in arrivals cause queues to grow rapidly, leading to a sharp rise in delay. As the cycle time increases, capacity improves and the risk of oversaturation decreases, causing this delay component to level out.

Performance Index (PI):

The PI graph shows how delay and stops affect the overall performance. For the base traffic flow, the optimal range is found at lower cycle times. Increasing the cycle time beyond what is necessary (e.g., to 120 or 150 seconds) results in a noticeably higher overall "cost" for road users.

4.2.3 Sensitivity analysis: Approaching capacity

To examine how the system performs under higher demand, the traffic volumes were increased.

115% Flow:

The network managed a 15% increase without major issues. The highest Degree of Saturation (DOS) rose to 85.85%, which is close to but still below the theoretical capacity limit of 90%.

130% Flow (Congested Scenario):

At a 30% increase, the system could no longer cope. The highest DOS reached 93.53%, indicating that the intersection was fully saturated. The total network Delay increased to 23.48 PCU-hr/hr. This confirms that the current intersection design is sensitive to large traffic increases, where even small disturbance would likely cause heavy queuing.

4.2.4 Impact of network modifications and cycle time adjustments

Two strategies were tested to manage the high demand (130% flow):

Adding a Third Stage: A separate stage for the Westbound left-turn was introduced to improve safety. However, this negatively impacted efficiency. The cycle time had to be increased to 130 seconds to compensate for the extra "lost time" between stages. Consequently, the Total Network Delay increased further to 37.7 PCU-hr/hr, and the Performance Index worsened to 562.21.

Fixed Cycle Times: Testing fixed cycle times (80s, 100s, 120s) under high demand scenarios (up to 135% flow) revealed a clear trade-off between capacity and delay. As shown in Table 3.4, increasing the cycle time successfully reduces the intersection's saturation. For the critical 135% flow scenario, increasing the cycle from 80s to 120s lowered the highest Degree of Saturation (DOS) from 89.46% to 86.25%, effectively keeping the intersection within capacity limits (<90%).

However, this gain in capacity comes at the direct cost of increased waiting times. For the same 135% flow, the Total Network Delay rose from 17.76 to 20.3 PCU-hr/hr as the cycle time was extended. This suggests that while longer cycle times provide a better safety margin against oversaturation, a shorter cycle (e.g., 80s) might be preferable as it maintains acceptable capacity (89.46% DOS) while minimizing the delay for road users.

4.2.5 Qualitative recommendations for improvement

Based on the heavy congestion observed in the oversaturated high-demand scenario (20–30% increase), it is clear that signal optimization alone is not enough. To improve traffic performance and increase capacity, the following qualitative measures are recommended:

Geometric Improvements (Channelization):

The most effective measure is to add separate turning lanes. Separating left-turning vehicles

from through-traffic prevents the through lanes from being blocked and reduces the risk of spillback.

Protected Signal Phases:

Introducing protected left-turn phases would remove conflicts with opposing traffic. Although the simulation of an additional signal stage showed higher delay due to extra lost time, this measure is important for safety. To make it efficient, it should be combined with the geometric widening mentioned above.

Signal Coordination:

If there are nearby intersections, coordinating the signals to create a Green Wave can help maintain smooth platoon flow. This ensures that vehicles leaving the intersection do not immediately face a red light further downstream, reducing the risk of queues forming back into the junction.

4.3 VISSIM

4.3.1 Performance Comparison with TRANSYT

Base scenario

The uniform vehicle delay estimated by TRANSYT is 22.2 seconds per vehicle, compared with 28.0 seconds per vehicle from the delay results and 24.1 seconds per vehicle reported in VISSIM's network performance. This discrepancy arises from the deterministic nature of TRANSYT versus the stochastic modeling in VISSIM. TRANSYT calculates uniform delay using analytical Webster-type formulas that rely on cycle length, effective green time, and arrival patterns. It assumes steady arrivals and ideal discharge conditions, so in light traffic many vehicles pass without stopping, resulting in relatively small delays. VISSIM, on the other hand, is a microscopic simulation that represents the behavior of individual vehicles. Even under light traffic, vehicles may slow down, yield, change lanes, or be influenced by pedestrians and random driver behavior. These micro-interactions generate additional delays that TRANSYT's simplified formulas do not account for.

Changed traffic conditions

When comparing vehicle delays between real-world traffic flow in VISSIM and increased traffic flow modeled in TRANSYT, it becomes evident that the delay predicted by VISSIM falls within the range of delays predicted by TRANSYT under conditions of traffic flow approaching capacity.

4.3.2 Attempted Improvement

Initial tests involving reduced cycle times and adjusted green splits produced only marginal improvements—on the order of one second. Given the inherent stochasticity of VISSIM, the true performance gain is likely even smaller. This suggests that while further optimization of cycle time based on real-world traffic patterns may still offer some benefit, the overall impact is expected to remain limited.

4.3.3 Performance Comparison between Fixed Timed Signal and Vehicle Actuated Control

Table 4.1 compares fixed signal timing with vehicle-actuated control in the base scenario. The results show that vehicle-actuated control reduces both average vehicle delay and average pedestrian delay by 5.2 seconds. This consistent reduction demonstrates that adaptive control enhances intersection performance without prioritizing one mode over the other, making it more effective than fixed-timed signals in balancing traffic and pedestrian flows.

Table 4.1 Delay comparison between fixed time signal and vehicle actuated control

	Average Vehicle Delay (s)	Average Pedestrian Delay (s)
Fixed Timed Signal	28.0	0
Vehicle Actuated Control	22.8	0